

16 March

nonlinear methods

Let's revisit Newton's method

$$Gm = d$$

we want to minimize

$$\phi(m) = \frac{1}{2} \|d - Gm\|^2$$

let's do Taylor expansion:

$$\begin{aligned}\phi(m_k + \Delta m) &\approx \phi(m_k) + \nabla \phi(m_k)^T \Delta m \\ &\quad + \frac{1}{2} \Delta m^T H(m_k) \Delta m\end{aligned}$$

$\nabla \phi$ is the gradient:

$$= \begin{bmatrix} \frac{\partial \phi}{\partial m_1} \\ \vdots \\ \frac{\partial \phi}{\partial m_m} \end{bmatrix}$$

H is the Hessian:

$$= \begin{bmatrix} \frac{\partial^2 \phi}{\partial m_1^2} & \dots & \frac{\partial^2 \phi}{\partial m_1 \partial m_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 \phi}{\partial m_m \partial m_1} & \dots & \frac{\partial^2 \phi}{\partial m_m^2} \end{bmatrix}$$

Let's take derivative wrt Δm :

$$\frac{\partial}{\partial \Delta m} (\phi(m_k + \Delta m)) = \nabla \phi(m_k) + H \Delta m$$

and set this to zero:

$$\nabla \phi(m_k) = -H \Delta m \quad \left[\text{see eqn 9.12 in Aster} \right]$$

Let's compute the gradient:

recall $\phi(m) = \frac{1}{2} r^T r = \frac{1}{2} \|r\|^2$

where $r = d - Gm$

take derivative of $\frac{1}{2} r^T r$:

$$\begin{aligned} \nabla \phi(m) &= \frac{\partial}{\partial m} \left(\frac{1}{2} r^T r \right) \\ &= \left(\frac{\partial r}{\partial m} \right)^T r \end{aligned}$$

now since $r = d - G(m)$

then $\frac{\partial r}{\partial m} = - \frac{\partial G}{\partial m}$

where Jacobian $J = \frac{\partial G}{\partial m}$

$$J = \begin{bmatrix} \frac{\partial g}{\partial m_1} & \dots & \dots \\ \vdots & \ddots & \vdots \end{bmatrix}$$

$\hookrightarrow \nabla \phi(m) = -J^T r$ [see eqn 9.19 in Aster]

recall Hessian $H = \nabla^2 \phi$

and since $\nabla \phi = -J^T r$

then $H = \nabla^2 \phi = J^T J - \underbrace{\sum_{i=1}^n r_i \frac{\partial^2 G_i}{\partial m^2}}_{= Q(m)}$

[see eqn 9.26]

recall $\nabla \phi(m_k) = -H \Delta m$

then $\Delta m = -H^{-1} \nabla \phi(m_k)$

substitute back in:

$$\Delta m = (J^T J - Q(m))^{-1} J^T r$$

by assuming $Q(m) \approx 0$,

we are left w/

Gauss-Newton (GN)

hence $H \approx J^T J$ [see eqn 9.28]

in GN, we ignore $Q(m)$
b/c it's probably small &
super expensive to compute.
But only works when near
the solution

Take eqns 9.12, 9.19, & 9.28:

$$J(m_k)^T J(m_k) \Delta m = J(m_k)^T F(m_k)$$

$$\rightarrow J^T J \Delta m = -J^T F$$

GN can fail converge, or
converge to local maxima
or local minima

For Levenberg-Marquardt :

$$(\mathbf{J}^T \mathbf{J} + \lambda \mathbf{I}) \Delta \mathbf{m} = -\mathbf{J}^T \mathbf{r}$$

Method	Hessian Used	Convergence	Cost
Gradient Descent	No	Slow	Cheap
Newton	Exact	Quadratic	Very expensive
Gauss-Newton	Approximate	<u>Superlinear</u> (near solution)	Moderate
Levenberg-Marquardt	Damped GN	More stable	Slightly higher